2 Real Analysis 2023

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Final Exam

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Student Number:									

Instructions:

- This question paper consists of **9 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the corresponding boxes.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 70 points = 100% (74 points available).

Student Number:									

Problem 1

(a) Consider the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^4$.

(i) Determine, without proof, the image $f(A)$ of the set $A = [-1, 2]$. (1 points)

Solution. $f(A) = [0, 16]$.

(ii) Determine, without proof, the pre-image $f^{-1}(B)$ of the set $B = [-1, 16]$. (1 points)

Solution. $f^{-1}(B) = [-2, 2]$.

(b) Consider the set $S = [0, 1] \cup \left\{ \frac{2n}{n+1} : n \in \mathbb{N} \right\} \cup \{5\}$.

(i) Determine, without proof, the set S' of all limit points of S. (1 points)

Solution. $S' = [0, 1] \cup \{2\}$.

(ii) Determine, without proof, the set of all isolated points of S. (1 points)

Solution. $\left\{ \frac{2n}{n+1} : n = 2, 3, 4, \ldots \right\} \cup \{5\}$.

(c) Give an example of two countable sets A and B such that their intersection $A \cap B$ is not countable. (2 points)

Solution. For example, $A = \{1, 3, 5, 7, \ldots\}$ and $B = \{2, 4, 6, 8, \ldots\}$. Both sets A and B are countable, but their intersection $A \cap B = \emptyset$ is not countable.

(d) Determine the intersection $\bigcap\limits_{n=1}^{\infty} \left[0, \frac{1}{n^2}\right]$. Provide a full justification for your answer. (4 points)

Solution. $\bigcap\limits_{n=1}^{\infty} \left[0, \frac{1}{n^2}\right] = \{0\}$.

Since $0 \in \left[0, \frac{1}{n^2}\right]$ for all $n \in \mathbb{N}$, we have that $0 \in \bigcap\limits_{n=1}^{\infty} \left[0, \frac{1}{n^2}\right]$. It remains to prove that 0 is the only element of the intersection.

1st solution. If $x < 0$, then $x \notin \left[0, \frac{1}{n^2}\right]$ for any $n \in \mathbb{N}$, and therefore $x \notin \bigcap\limits_{n=1}^{\infty} \left[0, \frac{1}{n^2}\right]$.

If $x > 0$, then by the Archimedean Principle, applied to the number $\frac{1}{\sqrt{x}} \in \mathbb{R}$, there exists $n_0 \in \mathbb{N}$ with $n_0 > \frac{1}{\sqrt{x_0}}$. This inequality is equivalent to $x > \frac{1}{n_0^2}$. It follows that $x \notin \left[0, \frac{1}{n_0^2}\right]$, and therefore $x \notin \bigcap\limits_{n=1}^{\infty} \left[0, \frac{1}{n^2}\right]$.

We conclude that $\bigcap\limits_{n=1}^{\infty} \left[0, \frac{1}{n^2}\right] = \{0\}$.

2nd solution. Let $x \in \bigcap\limits_{n=1}^{\infty} \left[0, \frac{1}{n^2}\right]$.

For any $n \in \mathbb{N}$ we have $0, x \in \left[0, \frac{1}{n^2}\right]$. Hence, $|0 - x| \leq \frac{1}{n^2}$. Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\frac{1}{N^2} < \varepsilon$. We obtain that $|0 - x| < \varepsilon$ for any $\varepsilon > 0$. This is only possible if $|0 - x| = 0$, i.e., $x = 0$.

Student Number:									

Problem 2

For each of the following series, determine whether it is absolutely convergent, conditionally convergent or divergent. Give a justification.

(a) $\displaystyle\sum_{n=0}^{\infty} \frac{\cos(2n)}{(2n)^2+1}$ (4 points)

Solution. For any $n \in \mathbb{N}$ we have

$$\left|\frac{\cos(2n)}{(2n)^2+1}\right| \le \frac{1}{4n^2+1} \le \frac{1}{4n^2}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges (this is a convergent p-series with $p = 2$), and by the Arithmetic Limit Theorem the series $\sum_{n=1}^{\infty} \frac{1}{4n^2}$ converges. By the Comparison Test, the series $\sum_{n=0}^{\infty} \left|\frac{\cos(2n)}{(2n)^2+1}\right|$ converges.

We conclude that the series $\sum_{n=0}^{\infty} \frac{\cos(2n)}{(2n)^2+1}$ converges absolutely.

(b) $\displaystyle\sum_{n=0}^{\infty} (-1)^n \frac{(n!)^2}{(2n)!}$ (4 points)

Solution. We apply the Ratio Test:

$$\begin{aligned} L = \lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} &= \lim_{n\to\infty} \frac{((n+1)!)^2}{(2n+2)!} \frac{(2n)!}{(n!)^2} = \lim_{n\to\infty} \frac{(n+1)(n+1)}{(2n+2)(2n+1)} \\ &= \lim_{n\to\infty} \frac{\left(1+\frac{1}{n}\right)\left(1+\frac{1}{n}\right)}{\left(2+\frac{2}{n}\right)\left(2+\frac{1}{n}\right)} = \frac{1}{4} < 1. \end{aligned}$$

We conclude that the series $\sum_{n=0}^{\infty} (-1)^n \frac{(n!)^2}{(2n)!}$ converges absolutely.

(c) $\displaystyle\sum_{n=1}^{\infty} (-1)^n \frac{n+1}{2n}$ (2 points)

Solution. We have

$$\lim_{n\to\infty} \frac{n+1}{2n} = \lim_{n\to\infty} \frac{1+\frac{1}{n}}{2} = \frac{1}{2}.$$

It follows that the terms of the series $(-1)^n \frac{n+1}{2n}$ do not converge to zero as $n \to \infty$. We conclude that the series $\sum_{n=1}^{\infty} (-1)^n \frac{n+1}{2n}$ diverges.

(d) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}+1}$ (6 points)

Solution. First we consider the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}+1}$. For any $n \in \mathbb{N}$ we have

$$\frac{1}{\sqrt[3]{n}+1} \ge \frac{1}{\sqrt[3]{n}+\sqrt[3]{n}} = \frac{1}{2n^{\frac{1}{3}}}.$$

The series $\sum_{n=1}^{\infty} \frac{1}{n^{\frac{1}{3}}}$ diverges (this is a divergent p-series with $p = \frac{1}{3}$). By the Arithmetic Limit Theorem, the series $\sum_{n=1}^{\infty} \frac{1}{2n^{\frac{1}{3}}}$ diverges. By the Comparison Test, the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}+1}$ diverges. Thus, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}+1}$ is not absolutely convergent.

The series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}+1}$ is an alternating series. We have that $\lim_{n\to\infty} \frac{1}{\sqrt[3]{n}+1} = 0$ and $\frac{1}{\sqrt[3]{n}+1} \ge \frac{1}{\sqrt[3]{n+1}+1}$, $n \in \mathbb{N}$. By the Alternating Series Test, the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}+1}$ converges.

The final answer is that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}+1}$ converges conditionally.

Student Number:									

Problem 3

Read the following theorem and an outline of its proof.

The Bolzano-Weierstrass Theorem. Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.

Proof. Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Let $m, M \in \mathbb{R}$ be such that $m \leq a_n \leq M$ for all $n \in \mathbb{N}$. We can choose $m < M$. Put $I_0 = [m, M]$. Let $\ell = M - m > 0$ be the length of I_0.

Using the midpoint of I_0, we divide I_0 into two closed and bounded intervals; at least one of these intervals contains infinitely many terms of $(a_n)_{n=1}^{\infty}$. Repeating the process, we construct a sequence of bounded closed intervals $(I_k)_{k=0}^{\infty}$ with the following properties:

1) $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$,

2) each I_k has length $\left(\frac{1}{2}\right)^k \ell$,

3) each I_k contains infinitely many terms of $(a_n)_{n=1}^{\infty}$.

Take $a \in \bigcap_{k=0}^{\infty} I_k$. Put $n_0 = 1$ and then for each $k \in \mathbb{N}$ choose $n_k \in \mathbb{N}$ such that $n_k > n_{k-1}$ and $a_{n_k} \in I_k$. Then $(a_{n_k})_{k=0}^{\infty}$ is a subsequence of $(a_n)_{n=1}^{\infty}$ that converges to a.

Answer the following questions:

(a) Why can we find $m, M \in \mathbb{R}$ such that $m \leq a_n \leq M$ for all $n \in \mathbb{N}$? (2 points)

Solution. This is exactly the definition of a bounded sequence: $(a_n)_{n=1}^{\infty}$ is bounded below, i.e. there exists $m \in \mathbb{R}$ such that $a_n \geq m$ for all $n \in \mathbb{N}$. Moreover, $(a_n)_{n=1}^{\infty}$ is bounded above, i.e. there exists $M \in \mathbb{R}$ such that $a_n \leq M$ for all $n \in \mathbb{N}$.

(b) Explain why at least one of the two halves of the interval I_0 contains infinitely many terms of $(a_n)_{n=1}^{\infty}$. (2 points)

Solution. Assume that each of the halves of I_0 contains only finitely many terms of $(a_n)_{n=1}^{\infty}$. Then also I_0 contains only finitely many terms of $(a_n)_{n=1}^{\infty}$. But by construction I_0 contains all terms of $(a_n)_{n=1}^{\infty}$, and they are infinitely many. The contradiction shows that the assumption was wrong and at least one of the halves of I_0 must contain infinitely many terms of $(a_n)_{n=1}^{\infty}$.

(c) When we take $a \in \bigcap_{k=0}^{\infty} I_k$, we use the fact that $\bigcap_{k=0}^{\infty} I_k$ is non-empty. Which theorem or property guarantees this? Name this theorem or property and state it clearly. (2 points)

Solution. This is the Nested Interval Property: If $(I_k)_{k=0}^{\infty}$ is a sequence of non-empty closed bounded intervals with $I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots$, then $\bigcap_{k=0}^{\infty} I_k \neq \emptyset$.

(d) Prove that $\lim_{k\to\infty} a_{n_k} = a$. (5 points)

Solution. By construction, both numbers a and a_{n_k} belong to the interval I_k of length $\left(\frac{1}{2}\right)^k \ell$. It follows that

$$|a_{n_k} - a| \leq \left(\frac{1}{2}\right)^k \ell.$$

Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $\left(\frac{1}{2}\right)^N \ell < \varepsilon$. Then for all $k \geq N$ we have

$$|a_{n_k} - a| \leq \left(\frac{1}{2}\right)^k \ell \leq \left(\frac{1}{2}\right)^N \ell < \varepsilon.$$

By the definition of the limit, $\lim_{k\to\infty} a_{n_k} = a$.

(e) Give an example of a bounded divergent sequence, and of a converging subsequence of your sequence. (No proofs required.) (2 points)

Solution. For example, consider the sequence $(a_n)_{n=1}^\infty$ with $a_n = (-1)^n$. This sequence is divergent. However, $a_{2n} = (-1)^{2n} = 1$ is a constant sequence and therefore convergent. The subsequence $(a_{2n})_{n=1}^\infty$ is a convergent subsequence of $(a_n)_{n=1}^\infty$.

(f) Now give an example of a sequence that does not have convergent subsequences. (No proofs required.) (2 points)

Solution. For example, consider the sequence $(a_n)_{n=1}^\infty$ with $a_n = n$. Any subsequence of this sequence is unbounded and consequently divergent. Thus, $(a_n)_{n=1}^\infty$ does not contain convergent subsequences.

Student Number:									

Problem 4
Let $f : \mathbb{R} \to \mathbb{R}$. Suppose that $\lim_{x\to -1} f(x) = 3$. Prove by definition that

$$\lim_{x\to -1} \frac{2f(x)+1}{f(x)+1} = \frac{7}{4}.$$

(8 points)

Solution.
We have
$$\left|\frac{2f(x)+1}{f(x)+1} - \frac{7}{4}\right| = \left|\frac{4(2f(x)+1)-7(f(x)+1)}{4(f(x)+1)}\right| = \frac{|f(x)-3|}{4|f(x)+1|}.$$
By the definition of the limit $\lim_{x\to -1} f(x) = 3$, there is $\delta_1 > 0$ such that

$$0 < |x+1| < \delta_1 \implies |f(x)-3| < 1.$$

For such x we have the following chain of equivalent inequalities: $|f(x)-3| < 1 \iff -1 < f(x)-3 < 1 \iff 2 < f(x) < 4 \iff 3 < f(x)+1 < 5$, the latter inequality implying $|f(x)+1| > 3$. Thus, if $0 < |x+1| < \delta_1$, then

$$\left|\frac{2f(x)+1}{f(x)+1} - \frac{7}{4}\right| = \frac{|f(x)-3|}{4|f(x)+1|} < \frac{|f(x)-3|}{12} < |f(x)-3|.$$

Given $\varepsilon > 0$, there is $\delta_\varepsilon > 0$ such that

$$0 < |x+1| < \delta_\varepsilon \implies |f(x)-3| < \varepsilon.$$

Put $\delta = \min\{\delta_1, \delta_\varepsilon\}$. Then

$$0 < |x+1| < \delta \implies \left|\frac{2f(x)+1}{f(x)+1} - \frac{7}{4}\right| < |f(x)-3| < \varepsilon.$$

By the definition of the limit, $\lim_{x\to -1} \frac{2f(x)+1}{f(x)+1} = \frac{7}{4}$.

Student Number:									

Problem 5

Consider the function $f : (0, \infty] \to \mathbb{R}$, $f(x) = \frac{1}{x}$.

(a) Construct two sequences $(x_n)_{n=1}^{\infty}$, $(y_n)_{n=1}^{\infty}$ on $(0, 1]$ such that $\lim_{n\to\infty}(x_n - y_n) = 0$, but for any $n \in \mathbb{N}$ we have $|f(x_n) - f(y_n)| \geq \varepsilon$ with some fixed $\varepsilon > 0$. (5 points)

Solution. Take $x_n = \frac{1}{n}$, $y_n = \frac{1}{n+1}$, $n \in \mathbb{N}$. Then

$$\lim_{n\to\infty}(x_n - y_n) = \lim_{n\to\infty}\left(\frac{1}{n} - \frac{1}{n+1}\right) = 0.$$

But $f(x_n) = n$, $f(y_n) = n + 1$, so that

$$|f(x_n) - f(y_n)| = |n - (n+1) = 1| \quad \text{for all } n \in \mathbb{N}.$$

(b) Is f uniformly continuous on $(0, 1]$? Justify your answer. (2 points)

Solution. f is not uniformly continuous on $(0, 1]$. This follows from (a): the existence of two sequences with described properties is a criterion for a function not being uniformly continuous.

(c) Is f uniformly continuous on $[1, 2]$? Justify your answer. (3 points)

Solution. f is continuous on $[1, 2]$ and the set $[1, 2]$ is compact. We know that any continuous function on a compact set is uniformly continuous on that set. It follows that f is continuous on $[1, 2]$.

Student Number:									

Problem 6

Consider a sequence of functions $(f_n)_{n=1}^{\infty}$, where $f_n : \mathbb{R} \to \mathbb{R}$, $f_n(x) = e^{-nx^2}$.

(a) Determine the pointwise limit of the sequence $(f_n)_{n=1}^{\infty}$ on $\mathbb{R}$. (5 points)

Solution. If $x = 0$, then $f_n(0) = 1$ for all $n \in \mathbb{N}$, and thus $\lim_{n\to\infty} f_n(0) = 1$.
If $x \neq 0$, then $0 < e^{-x^2} < 1$. It follows that

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} e^{-nx^2} = \lim_{n\to\infty} \left(e^{-x^2}\right)^n = 0.$$

Thus, the pointwise limit of the sequence $(f_n)_{n=1}^{\infty}$ is

$$\lim_{n\to\infty} f_n(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

(b) Does $(f_n)_{n=1}^{\infty}$ converge to f uniformly? Justify your answer. (3 points)

Solution. For each $n \in \mathbb{N}$, the function f_n is continuous on $\mathbb{R}$. However, the limit function $f = \lim_{n\to\infty} f_n$ is discontinuous on $\mathbb{R}$ (more precisely, it is discontinuous at 0). We know that the uniform limit of a sequence of continuous functions is continuous. It follows that $(f_n)_{n=1}^{\infty}$ does not converge to f uniformly.

Student Number:									

Problem 7

Consider the series $\sum_{n=1}^{\infty} \frac{x^n}{2n}$, $x \in \mathbb{R}$.

(a) Show that this series converges absolutely and uniformly on any interval of the form $[-r, r]$, where $0 < r < 1$. (5 points)

Solution. If $x \in [-r, r]$, where $r \in (0, 1)$, then

$$\left|\frac{x^n}{2n}\right| \leq \frac{r^n}{2n} = M_n,$$

and the series $\sum_{n=1}^{\infty} M_n$ converges. This can be seen, for example, by the Comparison Test: for all $n \in \mathbb{N}$ we have $\frac{r^n}{2n} \leq r^n$, and the geometric series $\sum_{n=1}^{\infty} r^n$ converges. Alternatively, the Ratio Test can be used:

$$L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to\infty} \frac{r^{n+1}}{2n+2} \frac{2n}{r^n} = \lim_{n\to\infty} r \frac{2n}{2n+2} = r < 1.$$

By the Weierstrass M-Test, the series $\sum_{n=1}^{\infty} \frac{x^n}{2n}$, $x \in \mathbb{R}$ converges absolutely and uniformly on $[-r, r]$.

(b) Does this series converge on the interval $[-1, 1]$? (2 points)

Solution. No. The series $\sum_{n=1}^{\infty} \frac{x^n}{2n}$ diverges at $x = 1$: for $x = 1$ it takes the form $\sum_{n=1}^{\infty} \frac{1}{2n}$ which is the divergent harmonic series.